■ FindMinimum

`FindMinimum[`f`, {`x`, `x_0`}]` searches for a local minimum in f, starting from the point x=x_0.

`FindMinimum` returns a list of the form `{`f_{min}`, {`x`->`x_{min}`}}`, where f_{min} is the minimum value of f found, and x_{min} is the value of x for which it is found. ■ `FindMinimum[`f`, {`x`, {`x_0`, `x_1`}}]` searches for a local minimum in f using x_0 and x_1 as the first two values of x. This form must be used if symbolic derivatives of f cannot be found.
■ `FindMinimum[`f`, {`x`, `*xstart*`, `*xmin*`, `*xmax*`}]` searches for a local minimum, stopping the search if x ever gets outside the range *xmin* to *xmax*. ■ `FindMinimum[`f`, {`x`, `x_0`}, {`y`, `y_0`}, `…`]` searches for a local minimum in a function of several variables. ■ `FindMinimum` works by following the path of steepest descent from each point that it reaches. The minima it finds are local, but not necessarily global, ones. ■ The following options can be given:

`AccuracyGoal`	`6`	the accuracy sought in the value of the function at the minimum
`Gradient`	`Automatic`	the list of gradient functions `{D[`f`, `x`], D[`f`, `y`], `…`}`
`MaxIterations`	`30`	maximum number of iterations used
`WorkingPrecision`	`Precision[1.]`	the number of digits used in internal computations

■ See page 481. ■ See also: `D`, `Fit`.

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